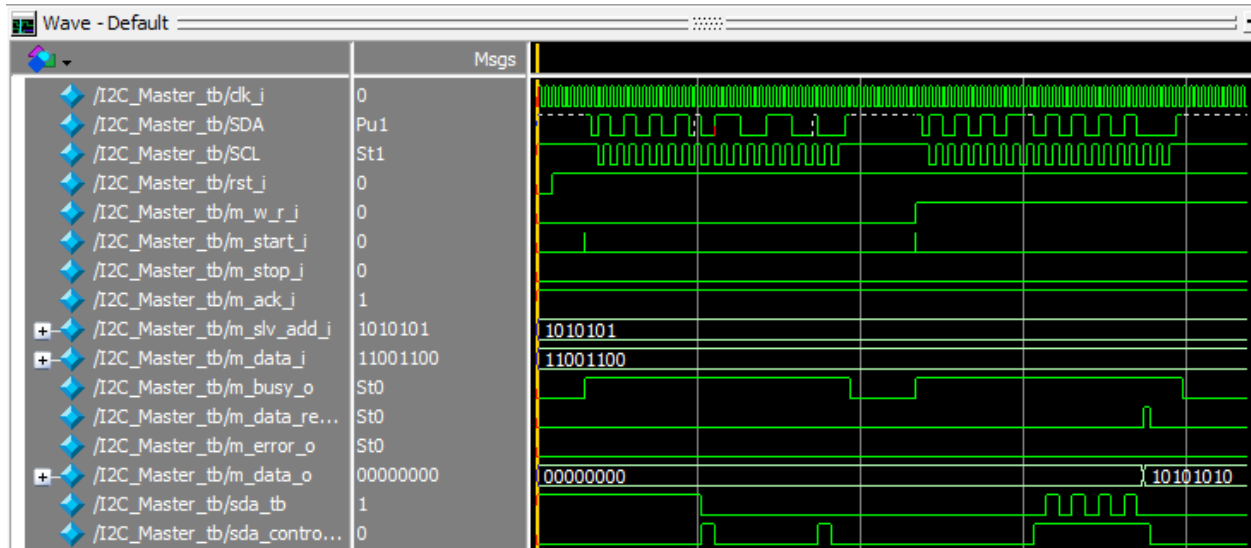
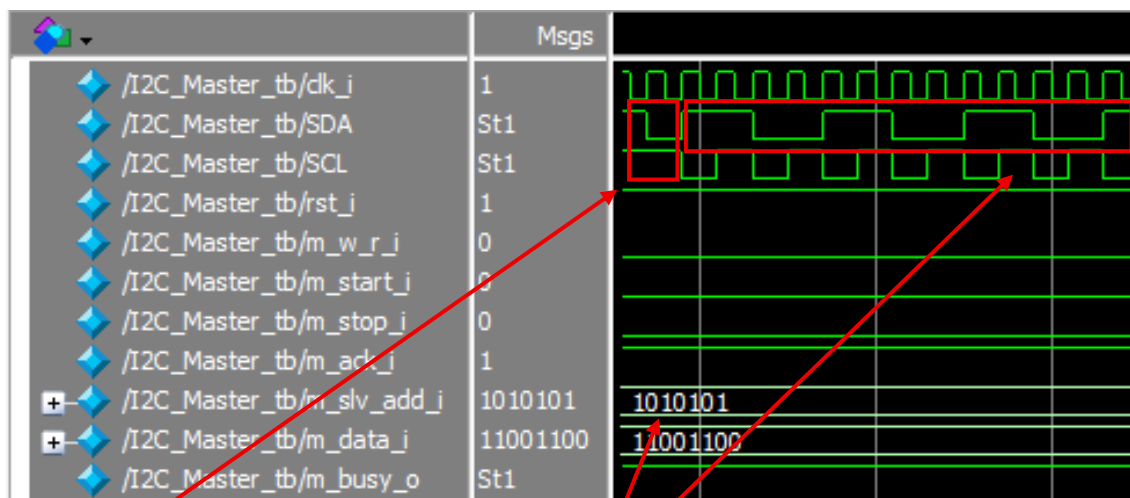
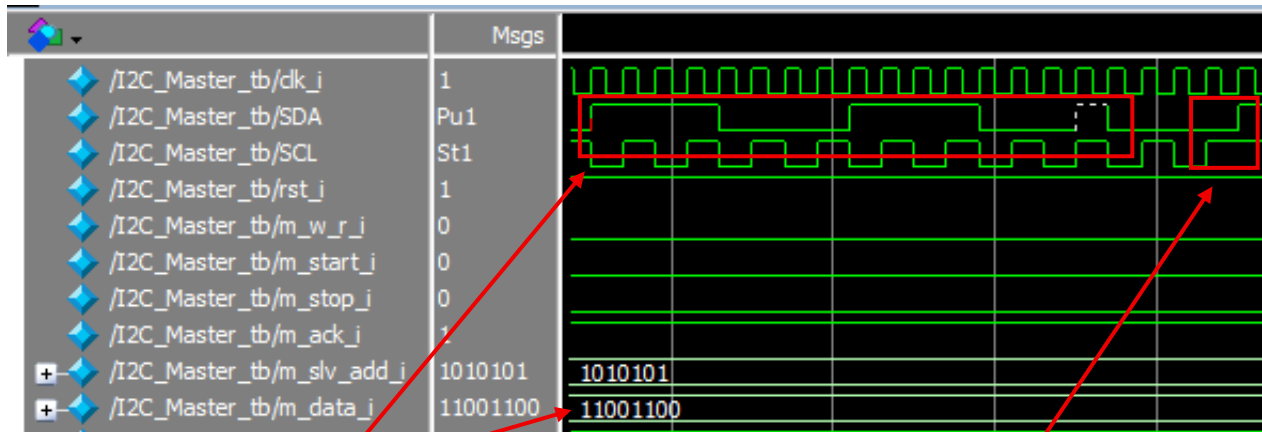


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Full test bench results:**Comments on the results (Testing Write):**

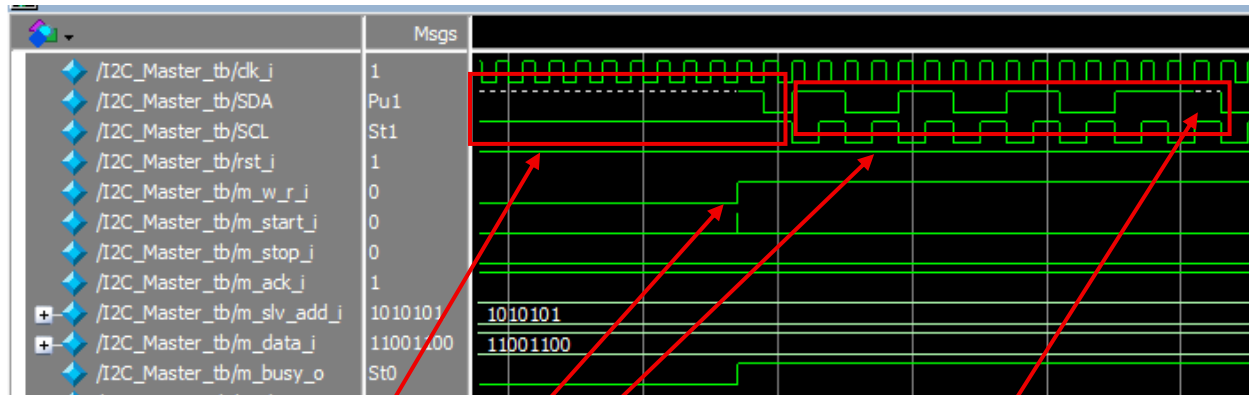
- Start condition, SDA goes from high to low while SCL is stable high
- Address is sent after starting condition (1010101)



- After sending address, data byte (11001100) is sent to SDA
- After sending data byte, SDA goes into high impedance state, indicating that slave is taking control of SDA to send acknowledgment bit
- Stop condition, SDA goes from low to high while SCL is stable high

```
# Starting WRITE test at time 150000
# Time: 150000 | State: 1 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 170000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 190000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 210000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 230000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 250000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 270000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 290000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 310000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 330000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 350000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 370000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 390000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 410000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 430000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 450000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 470000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 490000 | State: 3 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 510000 | State: 3 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 530000 | State: 4 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 550000 | State: 4 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 570000 | State: 4 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 590000 | State: 4 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 610000 | State: 4 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 630000 | State: 4 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 650000 | State: 4 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 670000 | State: 4 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 690000 | State: 4 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 710000 | State: 4 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 730000 | State: 4 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 750000 | State: 4 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 770000 | State: 4 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 790000 | State: 4 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 810000 | State: 4 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 830000 | State: 4 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 850000 | State: 6 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 870000 | State: 6 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 890000 | State: 7 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 910000 | State: 7 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 930000 | State: 7 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 950000 | State: 0 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# WRITE test complete at time 970000
```

Comment on the results (Testing Read):



- SDA goes into high impedance for Slave to detect address
- SDA sends Write bit (0 -> 1)
- Slave sends data byte on SDA
- SDA goes into high impedance state to wait for acknowledgment bit

Full Read results:

```
# Time: 970000 | State: 0 | SCL: 1 SDA: 1 | Busy: 0 | Data_Out: 00
# Starting READ test at time 1170000
# Time: 1170000 | State: 1 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1190000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1210000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1230000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1250000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1270000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1290000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1310000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1330000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1350000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1370000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1390000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1410000 | State: 2 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1430000 | State: 2 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1450000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1470000 | State: 2 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1490000 | State: 2 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1510000 | State: 3 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1530000 | State: 3 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1550000 | State: 5 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1570000 | State: 5 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1590000 | State: 5 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1610000 | State: 5 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1630000 | State: 5 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1650000 | State: 5 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1670000 | State: 5 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1690000 | State: 5 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1710000 | State: 5 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1730000 | State: 5 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1750000 | State: 5 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1770000 | State: 5 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1790000 | State: 5 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1810000 | State: 5 | SCL: 0 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1830000 | State: 5 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: 00
# Time: 1850000 | State: 5 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: 00
# Time: 1870000 | State: 6 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: aa
# Time: 1890000 | State: 6 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: aa
# Time: 1910000 | State: 7 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: aa
# Time: 1930000 | State: 7 | SCL: 0 SDA: 0 | Busy: 1 | Data_Out: aa
# Time: 1950000 | State: 7 | SCL: 1 SDA: 0 | Busy: 1 | Data_Out: aa
# Time: 1970000 | State: 0 | SCL: 1 SDA: 1 | Busy: 1 | Data_Out: aa
# READ test complete at time 1990000
# Time: 1990000 | State: 0 | SCL: 1 SDA: 1 | Busy: 0 | Data_Out: aa
# Note: $finish : C:/Users/Youss/Desktop/Verilog/Project/I2C_Master_tb.v(122)
# Time: 2190 ns Iteration: 0 Instance: /I2C_Master_tb
```